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भारतीय मानक मसौदा
बाल शल्य चिकित्सा के लिए मेज - विशिष्टि
[IS 8078 का पहला पुनरीक्षण]

Draft Indian Standard
Paediatric Operation Table - Specification
[First Revision of IS 8078]

ICS 11.140

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

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FOREWORD

(Formal clause will be added later)

This standard was originally published in 1976. The first revision of this standard has been brought out to align it with recent developments in the material requirements, removal of wooden material for handle and to bring the standard in line with the latest style and format of referred standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard specifies the construction and performance requirements for manually controlled paediatric operation table.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No.</i>	<i>Title</i>
IS 6911: 2026	Stainless Steel Plate, Sheet and Strip - Specification (<i>third revision</i>)
IS 17523: 2021	Castors and Wheels - Requirements for Castors for Furniture
IS 1068: 2025 ISO 1456: 2009	Metallic and other inorganic coatings - Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium (<i>fourth revision</i>)

3 TERMINOLOGY

3.1 Paediatric Operation Table - The specialized surgical table designed to support and position infant, child, and adolescent patients safely and comfortably during surgical procedures.

4 MATERIAL REQUIREMENTS

4.1 The materials used for the manufacture of the operation table shall be either corrosion-resistant or have a protective finish.

4.2 The raising column and the gearing system shall be made from materials that are wear-resistant during normal manipulation.

4.3 The handles provided shall be made from ebonite or plastics that can stand the action of mild acids.

4.4 The stainless-steel components used shall conform to designation X07Cr17Ni12Mo2 in accordance with IS 6911.

4.5 The anti-static rubber tread castors shall conform to IS 17523.

5 SHAPE AND DIMENSIONS

5.1 The shape of the operation table shall be as shown in Fig. 1

5.2 Table Top

The minimum overall length of the table top shall be 1500 mm. The overall width of the table top shall be 300 to 350 mm. The minimum length of head section and the leg section when drawn in shall be 220 and 400 mm respectively.

5.3 Height

The lowest position (without mattress) of the table top shall have a height of 750 ± 50 mm and at the extreme lifted position (without mattress) 1050 ± 50 mm.

6 CONSTRUCTION REQUIREMENTS

6.1 Base

The base shall be mounted on castors in such a way that it shall not render the operation table unstable. The base may have foot rails for providing comfortable foot rest to the surgical team. The base shall be fitted with a foot pedal to raise and lower the table top.

6.2 Table Top

The table top shall have minimum four sections that are head, trunk (two sections) and leg section. The head section shall be detachable. The detachable assembly of the leg section shall be optional. The entire table top shall have a smooth vertical movement. The table top shall satisfy the following requirements:

- The entire table top shall be capable of being tilted to $45 \pm 3^\circ$ Trendelenburg and reverse Trendelenburg positions;
- The table top shall have arrangements for a lateral tilt of $20 \pm 2^\circ$ in either direction;
- The table top may be provided with a perineal cut-out for drainage tray;
- The table top shall be able to obtain kidney position. If the position is not approachable, a separate kidney elevator shall be provided, which is capable of being raised to 150 mm from the table top and shall have a head-end control for manipulation;
- The entire table top may have a sweep of 360° in the horizontal plane;
- The upper half of the trunk section with the head section shall be capable of being raised to 90° and lowered to 30° .

6.2.1 Head and Leg Sections

The head section shall be capable of being raised to 40 to 45° and in reverse position 90° from the trunk portion. The leg section shall be capable of being lowered to at least 90° from the trunk to enable the table to be used as a chair and also for obtaining other positions.

6.3 Hydraulic Lifting Mechanism

The table shall have a hydraulic pump encased in the base for controlling the lift of the table and the pump shall be operated by a pedal. The table shall be lifted between 200 and 300 mm. The controls shall have easy access and various parts of the controls and the table shall be easy to clean and maintain. The parts intended to be removed by the user shall be easy to replace.

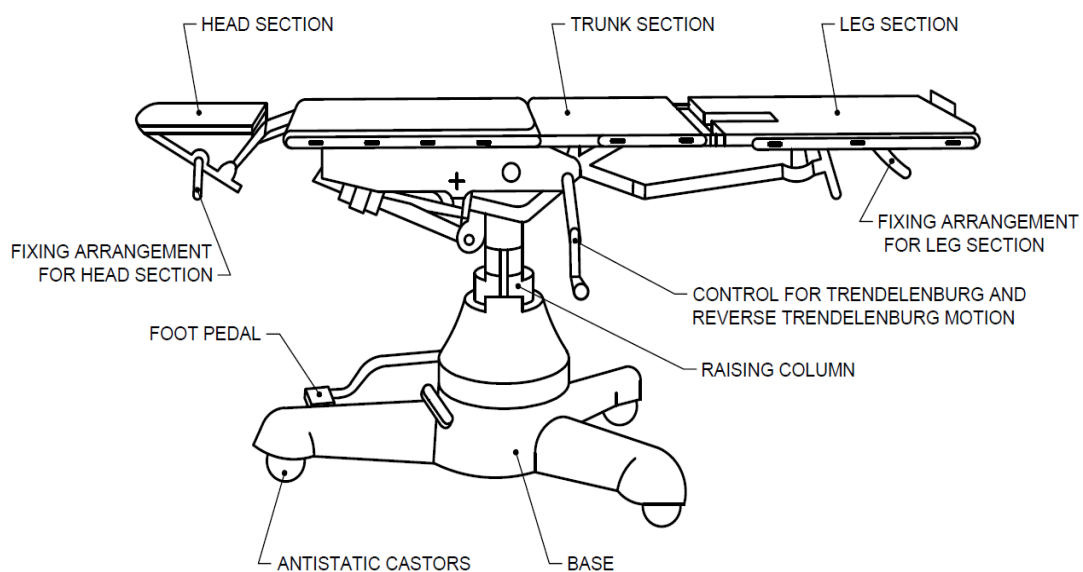


Fig. 1 PAEDIATRIC OPERATION TABLE

7 ACCESSORIES

7.1 The following standard accessories provided with the operation table:

TABLE 1 LIST OF ACCESSORIES

Sr No.	Accessories
1	Foot extension
2	Shoulder supports - side supports
3	Antistatic rubber pads
4	25 to 50 mm thick antistatic foam rubber mattress having electrical resistance between 10^4 and 10^6 ohms

7.2 The following accessories shall be optional:

TABLE 2 LIST OF ACCESSORIES

Sr No.	Accessories
1	Neck elevator or shoulder bridge
2	Permeable table top for radiography with cassette tray and necessary guide
3	Lateral cassette holder
4	Stainless steel transurethral tray with wire mesh screen
5	Instruments tray
6	Neuro-surgical attachment
7	Supports for pull-through operation

8 PERFORMANCE REQUIREMENTS

8.1 The table shall be stable on castors with floor locks.

8.2 The table shall be capable of moving on castors on a smooth cemented floor from its static position when a pull of maximum 100 N is applied, once the table is in movement 50 N pull shall be enough for movement.

8.3 The floor locks shall give rigidity to the table and it shall not move when a force of 500 N is applied to the base in any direction.

8.4 The force of 500 N is applied vertically on the extreme edge of the head section then the leg section shall not lift by more than 10 mm.

8.5 The mass of 5 kg is dropped from a height of 150 mm on the extreme end of the head section or centre section which is horizontal with table top and check for the table top damage.

8.6 The accessories are clamped to the table and checked for any wobble under normal working conditions.

8.7 Hydraulic Lifting Mechanism

The hydraulic lift shall give a vertical jerk-free and smooth movement to the system. When the hydraulic pressure is released, the table top shall take travel time of 15 seconds to 45 seconds from the maximum height to the bottom most position. The downward movement shall be smooth and jerk free. When the table top is held in any position by hydraulic action alone, and loaded with 100 kg in the trunk portion the height shall not reduce more than 3 mm in a period of four hours. The performance of the hydraulic lifting mechanism shall remain unimpaired with the table top with a mass of 100 kg kept on centre section when it is moved up and down 20 times.

9 WORKMANSHIP AND FINISH

9.1 The operation table components shall be arranged accurately to give smooth operation and additional performance requirements shall be as per 8.

9.2 The surfaces shall be well-finished and shall not have surface defects, such as pin-holes, blisters, blow-holes and similar other defects. The edges except those of the table top shall be rounded off.

9.3 The four sections comprising the table top in their normal position shall be in one plane when tested with a spirit level.

9.4 The welded joints shall have welds fully penetrating and well-finished.

9.5 The non-corrodible metal surface aluminium and stainless steel shall be matt-finished.

9.6 The corrosive surfaces like steel casting for base, cylinder castings and foot pedal shall be painted to non-reflecting shades which reflect the least such as sea green or any other shade as agreed to between the manufacturer purchaser.

9.7 The surface which to be painted shall be degreased and the rust-proofed by phosphating and then suitably protected by an anti-corrosive primer by brushing or spraying and then finished by spraying in stove enamel or air-drying enamel of the specified shade. The coatings of all instances shall be stoved or air-dried. The resulting finish shall be hard and shall not readily chip or flake off.

9.8 The handles and knobs shall have an easy grip and shall not impart colour or stain to the hand. The handles shall be of detachable type or folding type.

9.9 The operation table finish shall be hard and shall not readily chip or flake. The mild steel and brass parts shall be plated chromium over nickel according to **Service Condition 2** of IS 1068.

10 OPERATION MANUAL

The operation table shall be provided with instructions manual with the following information:

- a) Instructions for installation and maintenance (including spare parts);
- b) Instructions for correct operation of the table;
- c) Lift of the hydraulic lifting mechanism.

11 MARKING

11.1 The marking which is used by the manufacturer shall be legibly and indelibly marked with the Identification mark of the manufacturer.

11.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

12 PACKAGING

The operation table shall be packed as agreed to between the purchaser and the manufacturer. Each individual package shall contain only one unit and may be marked with the name of the item, and identification of the source of manufacture and batch number and any other marking if so desired.